

Class 1 SP250: Divisibility & Rainbow Method

This class may be the first math class you take outside your school. For most of you, this is the first **proof-based** class that you would be taking. This means that you will be required to make logical arguments to solve problems. This class is much more about **WHY** rather than **HOW**.

If a question asks “Is it possible”? The correct solution cannot be “It’s impossible because I tried it and it doesn’t work” because one can ask you to try harder.

Divisibility: Some starter questions.

- Is 75 divisible by 15?
- is 75 divisible by -5?
- is 50 divisible by $\frac{1}{2}$?
- Is 90 divisible by 4?

Divisibility (Definition): Let a and b be two integers with $b \neq 0$. We say that a is **divisible** by b (or b **divides** a , written as $b \mid a$) if there exists another integer c such that $a = b * c$.

- a) Is 75 divisible by 15? Yes because $75 = 5 * 15$. We write $15 \mid 75$.
- b) Is 75 divisible by -5? Yes because $75 = (-5) * (-15)$. We write $-5 \mid 75$.
- c) Is 50 divisible by $\frac{1}{2}$? No, because $\frac{1}{2}$ is not an integer.
- d) Is 90 divisible by 4? No because $90/4 = 45/2$ is not an integer.

Your first theorem:

Theorem: Let a, b, c be integers with $c \neq 0$.

- a) If both a and b are divisible by c (i.e. $c \mid a$ and $c \mid b$) $\Rightarrow c \mid (a+b)$.
- b) If a is divisible by c and b is not divisible by c , then $(a+b)$ is NOT divisible by c .
- c) If a is divisible by c and b is **any integer**, then $(a * b)$ is divisible by c .

Your first proof:

If both a and b are divisible by c (i.e. $c \mid a$ and $c \mid b$) $\Rightarrow c \mid (a+b)$.

First, **not a proof:** Say a is 30, b is 20 and c is 5. Then $5 \mid 30$ and $5 \mid 20$ and $5 \mid 50$. Done.

Proof v. example: This is an example and NOT a proof.

Example: for the **particular choices** of a, b, c satisfying the conditions, the statement is true.

Proof: For **all choices** of a, b, c satisfying the conditions, the statement is true!

“Fake Theorem”: All Math-m-addicts teachers have first names that are either 5 or 6 letters long.

Example: Tanya, Yirong, Sergey,

Unless you go through ALL the teachers and count the letters, you will not know whether this is true or not!

Counter-example: Madalena.

Back to the proof:

- a) If both a and b are divisible by c (i.e. $c \mid a$ and $c \mid b$) $\Rightarrow c \mid (a+b)$.

What are we given? We are given that $c \mid a$ and $c \mid b$. How do we put that “in math terms”?

We use the definition.

"a is divisible by c", so there exists an integer d such that $a = c * d$.

"b is divisible by c" so there exists an integer f such that $b = c * f$.

Now add these two equations.

$$a = c * d, b = c * f \Rightarrow (a+b) = cd + cf = c * (d+f).$$

Since d and f are both integers, so is (d+f). Hence, by definition, **(a+b)** is divisible by **c**, as required.

By subtracting rather than adding above, we also get that (a-b) is divisible by c if both a and b are.

b) If a is divisible by c and b is not divisible by c, then (a+b) is NOT divisible by c.

Question: What's needed to prove that m is divisible by n?

Answer: We write $m = n * p$ for some integer p. ← seems manageable.

Question: What's needed to prove that m is NOT divisible by n?

Answer: Need to show that there is no integer p such that $m = n * p$. ← no clue how to do this.

Logical aside: A theorem is a statement of the form: **If X then Y.**

X = "a is divisible by c and b is not divisible by c"

Y is "(a+b) is NOT divisible by c".

Sometimes, it is much easier to prove the following: **If (not Y), then (not X).**

This may seem trivial but can make a huge difference in many proofs. This is called "proof by contradiction".

Proof by contradiction:

Not Y: Assume that (a+b) is divisible by c.

Not X: it is **NOT** the case that a is divisible by c and b is NOT divisible by c.

If a is divisible by c, then by previous part of the theorem:

$b = (a+b) - a$ ← difference of two numbers divisible by c and hence is divisible by c.

Hence either a is not divisible by c or b is divisible by c, which is what we needed to prove.

c) If a is divisible by c and b is an integer, then (a*b) is divisible by c. ← Homework.

Problem 1. Tanya came into the store and bought 6 apples, 9 oranges and 21 mandarins. There is no sales tax. The cashier told her that Tanya's total is \$56. Tanya asked him to double check the bill. How did Tanya know that the total was wrong?

Solution: Since 6, 9 and 21 are all divisible by 3, the amount of money paid for apples, oranges and mandarins is a multiple of 3. By previous theorem, the total amount of money Tanya owes is a multiple of 3 and thus can not be \$56.

Rainbow Method:

What is the sum of numbers from 1 to 100?

$$1 + 2 + 3 + \cdots + 99 + 100 = X.$$

$$100 + 99 + 98 + \cdots + 2 + 1 = X.$$

Add these up:

$$100 \cdot 101 = 2X \implies X = \frac{100 \cdot 101}{2}.$$

What is the sum of integers from 1 to N?

$$1 + 2 + 3 + \cdots + N = X.$$

$$N + \cdots + 2 + 1 = X.$$

Add these up:

$$N \cdot (N + 1) = 2X \implies X = \frac{N \cdot (N + 1)}{2}.$$

In-class problem 2: What is the sum of all positive multiples of 3 less than 100?

Answer: $3 + 6 + 9 + \dots + 99 = 3(1 + 2 + \dots + 33) = 3 \cdot 33 \cdot 34 / 2 = 99 \cdot 17 = 1683$

What is the 100th even positive integer?

Answer: 200

What is the 100th odd positive integer?

Answer: It's 1 less than 100th even positive integer -- 199.

What's the 100th positive integer divisible by 4?

Answer: 400.

In-class problem 3: What's 100th positive integer NOT divisible by 4?

Answer: Every 4 integers, there are 3 non-multiples of 4 and one multiple of 4.

$100 = 3 \cdot 33 + 1$. ← 100th non-multiple of 4 is the 1st number in the 34'th block of 4.

The 33rd block ends on $33 \cdot 4 = 132$. Hence the number we seek is 133.

In-class problem 4: What's the sum of first 100 positive integers not divisible by 4?

Answer: $1 + 2 + 3 + \dots + 131 + 133$. ← 100 numbers on here.

It is tempting to say the sum is $50 \cdot (133 + 1)$. **However, this is WRONG.**

The rainbow method, in general, requires you to pair numbers up so that the sum in each pair is THE SAME!

If you remove 133, then the sum in each pair IS 132. Hence, the sum is $132 \cdot 99 / 2 + 133 = 6667$.